

# Clinical Risk Assessment Report: Patient 48

## 1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 51.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.32x the diagnostic threshold

### Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET), which elevated the patient's absolute risk by +8.2%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

## 2. Key Pathological & Imaging Drivers (Elevating Risk)

**Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET)**      **Value: 2.14 [Top 5%] (+8.2%)**  
*Points to continuous, elongated bands of high-density or highly active tumor tissue.*

**Large Zone Emphasis (CT) (GLZLM\_LGZE.CT)**      **Value: -0.33 [Bottom 19%] (+5.6%)**  
*Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.*

**75th Percentile Tumor Density (CONVENTIONAL\_HUQ3)**      **Value: 0.82 [Top 21%] (+3.5%)**  
*Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.*

**Small Zone Emphasis (PET) (GLZLM\_SZE.PET)**      **Value: 1.26 [Top 10%] (+3.1%)**  
*Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.*

**Large Zone Low Gray-Level Emphasis (CT) (GLZLM\_LZLGE.CT)**      **Value: 0.0 [Top 11%] (+2.9%)**  
*Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.*

**Maximum Tumor Density (CONVENTIONAL\_HUmax)**      **Value: 0.81 [Top 12%] (+2.8%)**  
*Reflects the most dense solid components or calcifications within the primary tumor lesion.*

**Short Run Low Gray-Level Emphasis (CT) (GLRLM\_SRLGE.CT)**      **Value: -0.23 [Bottom 9%] (+2.6%)**  
*Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.*

**Short Run High Gray-Level Emphasis (CT) (GLRLM\_SRHGE.CT)**      **Value: 0.72 [Avg (72th %ile)] (+2.2%)**  
*Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.*

## 3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

# Machine Learning Feature Attribution (SHAP Decision Path)

